

Drugs

Question Paper 2

Level	IGCSE
Subject	Biology
Exam Board	CIE
Topic	Drugs
Paper Type	(Extended) Theory Paper
Booklet	Question Paper 2

Time Allowed: 40 minutes

Score: /33

Percentage: /100

- 1 The fungus, *Trichophyton violaceum*, reproduces asexually by releasing spores.

A single spore was placed in the centre of a Petri dish containing an agar medium with starch and protein.

Fig. 1.1 shows the fungus that grew from the spore.

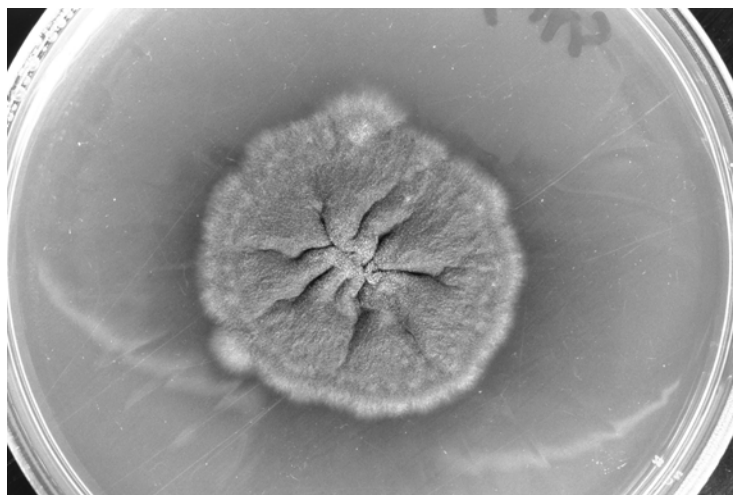


Fig. 1.1

- (a) State the name given to

- (i) the body of the fungus that grows from a single spore

..... [1]

- (ii) the thin threads that make up the body of the fungus.

..... [1]

- (b) Describe how a fungus, such as *T. violaceum*, obtains nutrients from the agar medium.

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Streptomyces, a soil bacterium, is a major source of antibiotics that are produced by pharmaceutical companies. An antibiotic sensitivity test can be carried out to help doctors decide which antibiotic should be used to treat a specific disease, such as gonorrhoea.

Gonorrhoea bacteria isolated from a person are grown on an agar medium. A ring with eight different antibiotics (1 to 8) is placed on the agar medium and left for 24 hours at 35 °C.

Fig. 1.2 shows the growth of bacteria on the agar medium after 24 hours.

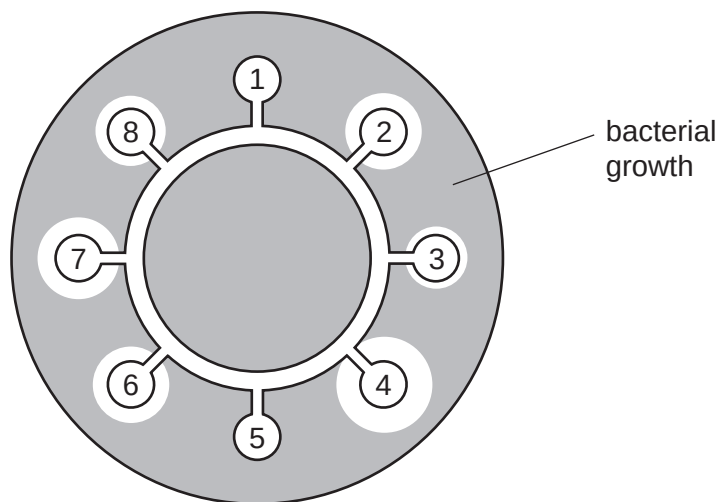


Fig. 1.2

(c) (i) Suggest why there is a clear area around some of the antibiotics.

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..... [2]

(ii) Explain why antibiotics 1 and 5 would **not** be chosen to treat the gonorrhoea infection.

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- (iii) People who take antibiotics should always be told to complete the treatment rather than stop taking the antibiotics when they feel better.

Suggest why people are given this advice.

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[Total: 12]

- 2 When bacteria are spread onto agar in a Petri dish they form colonies. Each colony forms from one bacterium. Fig. 4.1 shows an investigation into antibiotic resistance in a species of bacterium that causes disease.

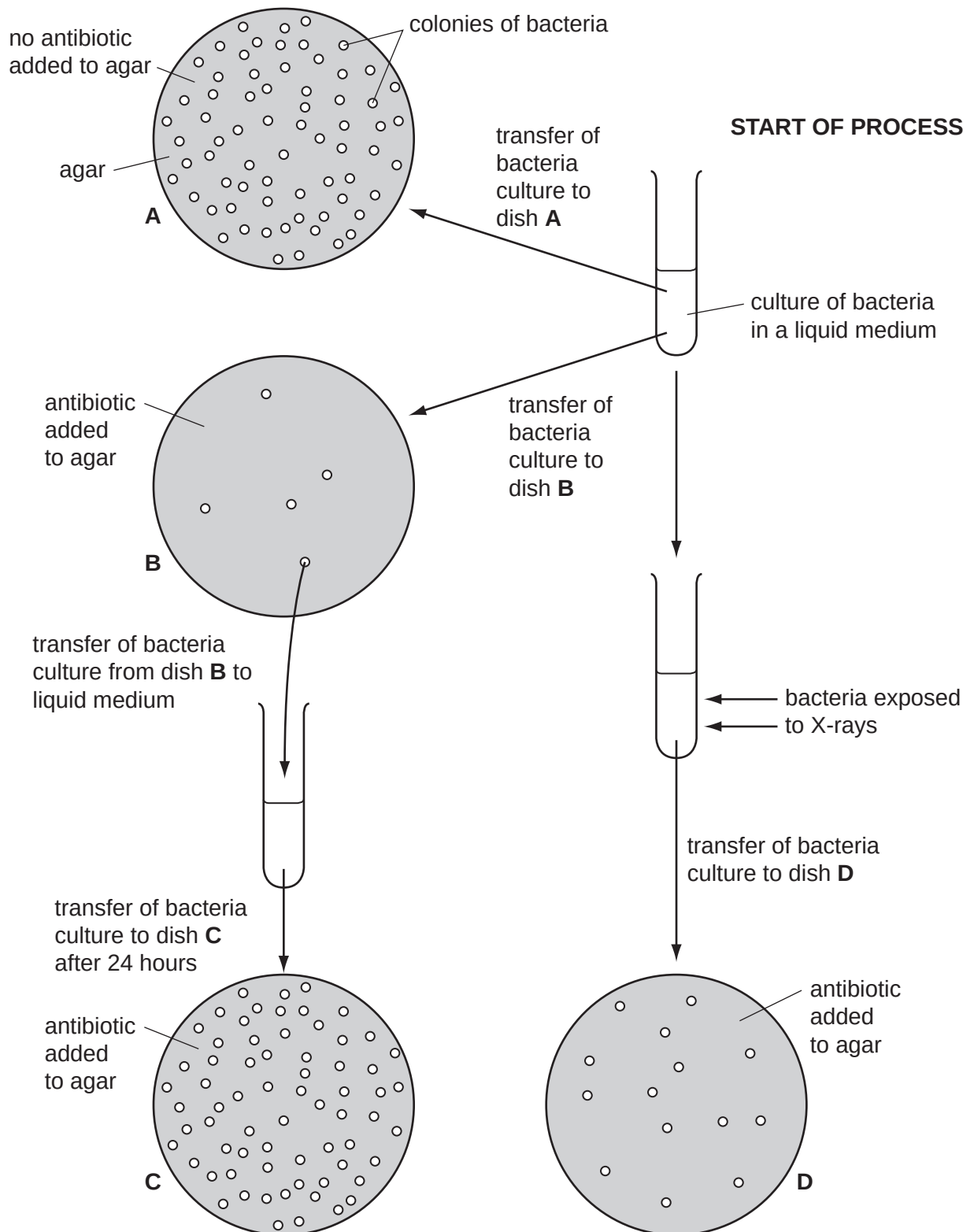


Fig. 4.1

(a) Explain what is meant by the term *antibiotic*.

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..... [2]

(b) Explain why

(i) only a few bacteria grew in dish **B** compared with dish **A**,

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..... [1]

(ii) more bacteria grew in **C** than in **B**.

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..... [1]

(c) Fig. 4.1 shows the effect of an antibiotic on a species of disease-causing bacterium.

Suggest why antibiotics should not be used too often.

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..... [2]

(d) Explain the possible effect of the X-rays on the bacteria.

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..... [3]

- (e) State two ways in which the **structure** of a bacterium differs from the **structure** of a virus.

1.
2. [2]

- (f) Human Immunodeficiency Virus (HIV) infects cells of the immune system.

Describe the effects of HIV on the immune system.

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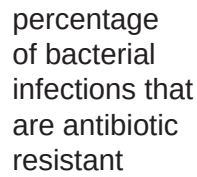
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[Total: 15]



[3]

(b) Many different antibiotics are used.

Suggest why some antibiotics are used less frequently than others.

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[Total: 6]